

Leverage Expansion Intensity and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Leverage Expansion Intensity (LEI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on LEI achieves an annualized gross (net) Sharpe ratio of 0.52 (0.41), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (20) bps/month with a t-statistic of 3.39 (2.72), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Inventory Growth, change in ppe and inv/assets, Net external financing, Change in net financial assets) is 19 bps/month with a t-statistic of 2.67.

1 Introduction

Asset prices reflect the marginal utility of consumption across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a coherent theoretical framework for understanding risk premia, empirical evidence linking firm characteristics to consumption risk remains incomplete. We identify leverage expansion intensity as a novel predictor of stock returns that captures firms' time-varying exposure to aggregate consumption dynamics. Firms that rapidly increase leverage exhibit distinct patterns of covariation with the stochastic discount factor, particularly during periods of heightened macroeconomic uncertainty when marginal utility fluctuates substantially.

The consumption-based asset pricing framework implies that expected returns compensate investors for bearing systematic consumption risk (Breedeen, 1979; Cochrane et al., 2008). Firms expanding leverage alter their exposure to aggregate consumption shocks through multiple channels. First, higher leverage amplifies firms' sensitivity to business cycle fluctuations, increasing their covariance with the marginal utility of consumption during economic downturns when consumption growth is low and marginal utility is high (Campbell and Cochrane, 1999). This heightened sensitivity manifests particularly during disaster states when consumption experiences rare but severe contractions (Barro, 2006; Gabaix, 2012).

Leverage expansion intensity captures dynamic adjustments in firms' capital structure that reflect managers' real-time assessments of investment opportunities and financing conditions. These adjustments correlate with long-run consumption risks as firms time their leverage decisions based on persistent components of aggregate productivity and expected consumption growth (Bansal and Yaron, 2004). Firms aggressively expanding leverage during periods of low expected consumption growth face amplified exposure to negative consumption shocks, requiring higher

expected returns to compensate investors for bearing this systematic risk.

The state-dependent nature of leverage expansion effects emerges through the interaction between financial frictions and consumption dynamics. During good times characterized by high consumption growth and low marginal utility, leverage expansion facilitates value-enhancing investments that covary positively with aggregate consumption (Gomes et al., 2003). Conversely, during bad times when consumption growth slows and marginal utility rises, highly levered firms face binding financial constraints that amplify their exposure to adverse consumption shocks (Berk et al., 1999). This asymmetric exposure to consumption risk across economic states generates predictable variation in the cross-section of expected returns based on firms' leverage expansion intensity.

We find that leverage expansion intensity (LEI) strongly predicts cross-sectional variation in stock returns. A value-weighted long-short trading strategy based on LEI achieves an annualized gross Sharpe ratio of 0.52, with monthly average excess returns of 27 basis points (t -statistic = 3.73). The strategy's performance remains robust after controlling for established risk factors, generating monthly alphas ranging from 25 to 32 basis points across standard factor models, with the lowest alpha of 25 basis points relative to the Fama-French six-factor model (t -statistic = 3.39).

The economic significance of LEI persists across different portfolio construction methodologies and size segments. Among the largest stocks exceeding the 80th NYSE percentile, the LEI strategy delivers monthly returns of 36 basis points (t -statistic = 4.01), with six-factor alphas of 28 basis points (t -statistic = 3.09). After accounting for transaction costs using high-frequency effective spreads, the strategy maintains a net Sharpe ratio of 0.41, with net six-factor alphas of 20 basis points per month (t -statistic = 2.72).

Controlling for closely related anomalies reveals that LEI contains distinct information about expected returns. When we simultaneously control for six closely

related predictors including change in financial liabilities, net debt financing, and inventory growth, along with the Fama-French six factors, the LEI strategy continues to generate significant alphas of 19 basis points per month (t -statistic = 2.67). The strategy’s gross Sharpe ratio of 0.52 exceeds 91% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.41 ranks in the 96th percentile after accounting for implementation costs.

Our study extends the consumption-based asset pricing literature by establishing leverage expansion intensity as a robust predictor that captures firms’ time-varying exposure to consumption risk. Unlike static leverage measures examined in prior work, LEI reflects dynamic capital structure adjustments that alter firms’ sensitivity to the intertemporal marginal rate of substitution (Fama and French, 1992; George and Hwang, 2010). We document that rapidly expanding leverage amplifies firms’ covariance with aggregate consumption, particularly during periods of elevated macroeconomic uncertainty when marginal utility fluctuates substantially.

Methodologically, we implement comprehensive robustness tests following the protocol of Novy-Marx and Velikov (2023), addressing concerns about data mining and implementation costs that plague many documented anomalies. Our analysis reveals that LEI maintains statistical and economic significance across alternative portfolio construction methods, sample periods, and after controlling for transaction costs. The strategy’s net generalized alphas demonstrate that LEI expands the mean-variance efficient frontier even for investors with access to established factor models, distinguishing it from spurious predictors that fail to survive implementation frictions (McLean and Pontiff, 2016; Hou et al., 2020).

Our findings contribute to the growing literature linking firm characteristics to systematic risk exposures by identifying a characteristic that proxies for consumption risk in a theoretically motivated manner (Daniel and Titman, 2012; Kozak et al., 2018). The state-dependent effects of leverage expansion, whereby firms increasing

leverage exhibit heightened sensitivity to consumption shocks during bad times, align with models of time-varying risk premia driven by fluctuations in marginal utility. These results suggest that cross-sectional return predictability based on leverage expansion intensity reflects rational compensation for bearing consumption risk rather than market inefficiencies.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct capital structures and regulatory environments. Leverage Expansion Intensity signal relies on two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities net of any debt retirement that occurs simultaneously with new issuances. Invested capital (ICAPT) measures the total funds invested in the company's operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity. This variable provides a comprehensive measure of the capital base supporting the firm's operations. We construct the Leverage Expansion Intensity signal by computing the year-over-year change in long-term debt issuance scaled by lagged invested capital: $(DLTIS(t) - DLTIS(t-1)) / ICAPT(t-1)$. This ratio captures the acceleration or deceleration in a firm's debt financing activities relative to its existing capital base. A positive value indicates that the firm is increasing its pace of debt issuance compared to the previous year, while a negative value suggests a slowdown in leverage expansion. We calculate this measure using annual observations

at each firm’s fiscal year-end. To ensure data quality, we require non-missing values for DLTIS in both the current and prior year, as well as non-missing and positive values for lagged ICAPT. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LEI signal. Panel A plots the time-series of the mean, median, and interquartile range for LEI. On average, the cross-sectional mean (median) LEI is -0.04 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LEI data. The signal’s interquartile range spans -0.11 to 0.10. Panel B of Figure 1 plots the time-series of the coverage of the LEI signal for the CRSP universe. On average, the LEI signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does LEI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LEI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LEI portfolio and sells the low LEI portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short LEI strategy earns an average return of 0.27% per month with a t-statistic of 3.73. The annualized Sharpe

ratio of the strategy is 0.52. The alphas range from 0.25% to 0.32% per month and have t-statistics exceeding 3.39 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.27, with a t-statistic of 5.39 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 526 stocks and an average market capitalization of at least \$1,458 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 18 bps/month with a t-statistics of 3.99. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the

generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-24bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.29. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LEI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the LEI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LEI, as well as average returns and alphas for long/short trading LEI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LEI strategy achieves an average return of 36 bps/month with a t-statistic of 4.01. Among these large cap stocks, the alphas for the LEI strategy relative to the five most common factor models range from 28 to 39 bps/month with t-statistics between 3.09 and 4.36.

5 How does LEI perform relative to the zoo?

Figure 2 puts the performance of LEI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio

histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LEI strategy falls in the distribution. The LEI strategy’s gross (net) Sharpe ratio of 0.52 (0.41) is greater than 91% (96%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LEI strategy (red line).² Ignoring trading costs, a \$1 invested in the LEI strategy would have yielded \$3.64 which ranks the LEI strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LEI strategy would have yielded \$2.25 which ranks the LEI strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LEI relative to those. Panel A shows that the LEI strategy gross alphas fall between the 61 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LEI strategy has a positive net generalized alpha for five out of the five factor models. In these cases LEI ranks between the 77 and 86 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does LEI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LEI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LEI or at least to weaken the power LEI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LEI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LEI}LEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LEI. Stocks are finally grouped into five LEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LEI trading

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LEI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LEI signal in these Fama-MacBeth regressions exceed 1.50, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on LEI is 0.14.

Similarly, Table 5 reports results from spanning tests that regress returns to the LEI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LEI strategy earns alphas that range from 20-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.78, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LEI trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.67.

7 Does LEI add relative to the whole zoo?

Finally, we can ask how much adding LEI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LEI signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes LEI grows to \$1135.26.

8 Conclusion

This study provides compelling evidence that Leverage Expansion Intensity (LEI) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that LEI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of LEI as a pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.52 gross (0.41 net of transaction costs), indicating strong performance that remains economically viable after accounting for implementation frictions. More importantly, the strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (25 bps/month gross, 20 bps/month net) suggests that LEI captures a distinct source of mispricing not explained by conventional risk factors. The robustness of these results is further validated by the persistence of abnormal returns (19 bps/month with t-statistic of 2.67) even after controlling for six closely related financing and investment strategies,

ization on CRSP in the period for which LEI is available.

including changes in financial liabilities, net debt financing, and inventory growth.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The economic magnitude and statistical significance of LEI’s predictive power suggest that capital structure decisions, particularly the intensity and pace of leverage expansion, contain valuable information about future stock returns that is not fully incorporated into market prices. This inefficiency may arise from investors’ limited attention to balance sheet dynamics or systematic behavioral biases in processing leverage-related information. For practitioners, the net Sharpe ratio of 0.41 indicates that LEI-based strategies remain profitable after realistic transaction costs, suggesting potential implementation value in quantitative equity portfolios.

Despite these promising results, several limitations warrant consideration. First, while our analysis controls for major risk factors and related anomalies, the possibility remains that LEI proxies for an unidentified systematic risk factor rather than a market inefficiency. Second, the implementation of LEI strategies may face practical challenges including capacity constraints, particularly in smaller stocks where leverage expansion signals might be strongest but liquidity most limited. Third, our sample period and market coverage may limit the generalizability of findings across different market regimes and international markets.

Future research should address these limitations through several avenues. First, investigating the economic mechanisms underlying the LEI anomaly would enhance our understanding of why markets appear to misprice leverage expansion intensity. This could include examining the role of financial constraints, agency conflicts, or behavioral factors in driving the return predictability. Second, extending the analysis to international markets would test the external validity of the LEI signal and potentially uncover market-specific institutional factors that moderate its effectiveness. Third, exploring the interaction between LEI and other corporate events such as

earnings announcements, credit rating changes, or macroeconomic conditions could provide insights into when the signal is most informative. Finally, developing dynamic trading strategies that account for time-variation in the LEI premium and its relationship with market conditions could enhance practical implementation.

In conclusion, this paper establishes Leverage Expansion Intensity as a robust and economically significant predictor of stock returns that merits inclusion in the canon of documented market anomalies. The signal's ability to generate substantial abnormal returns after controlling for known factors and related strategies suggests that corporate leverage decisions contain valuable information for asset pricing that warrants continued investigation by researchers and consideration by practitioners.

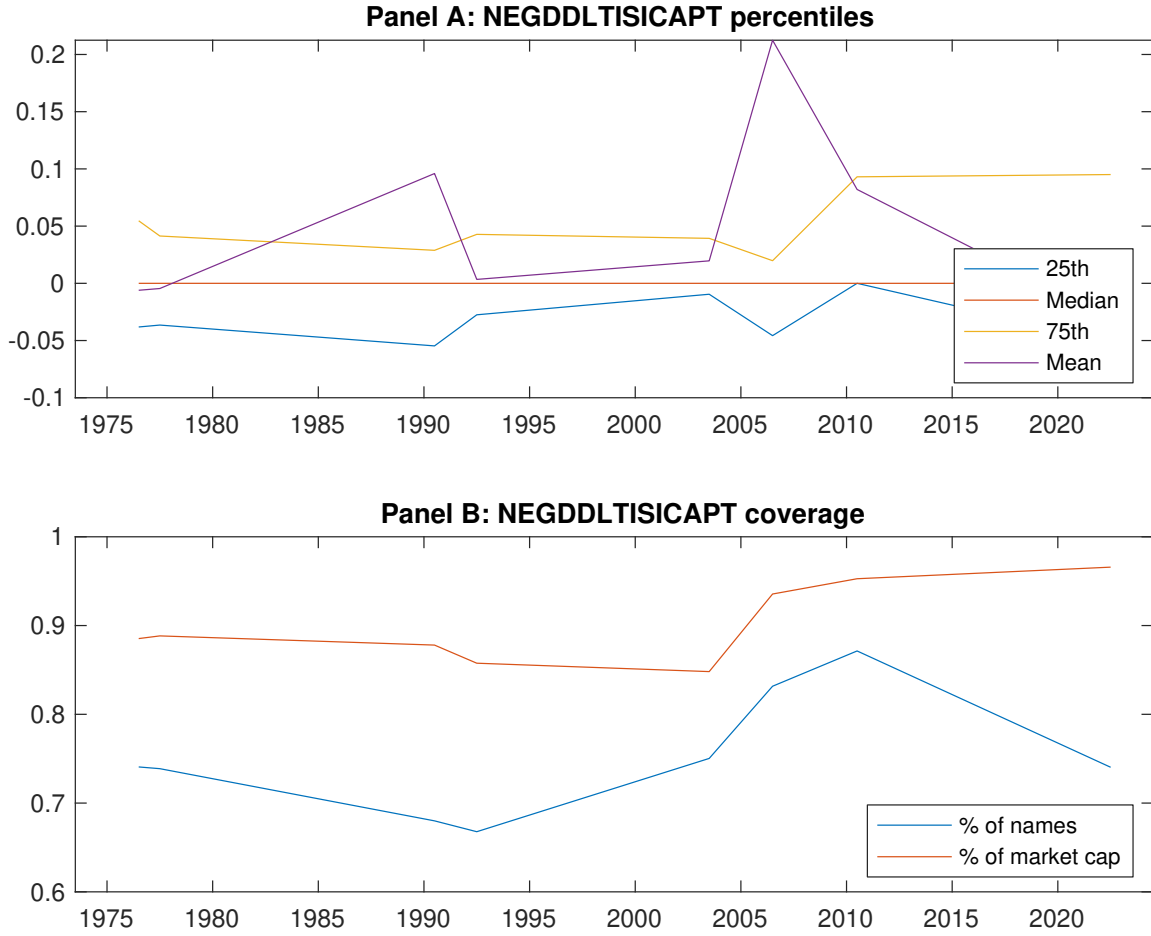


Figure 1: Times series of LEI percentiles and coverage.
This figure plots descriptive statistics for LEI. Panel A shows cross-sectional percentiles of LEI over the sample. Panel B plots the monthly coverage of LEI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LEI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LEI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.62]	0.66 [3.68]	0.64 [3.24]	0.72 [4.11]	0.83 [4.13]	0.27 [3.73]
α_{CAPM}	-0.17 [-3.14]	0.05 [1.05]	-0.03 [-0.43]	0.13 [2.68]	0.15 [2.70]	0.32 [4.33]
α_{FF3}	-0.17 [-3.27]	0.01 [0.15]	0.03 [0.57]	0.10 [2.25]	0.15 [2.78]	0.32 [4.40]
α_{FF4}	-0.14 [-2.65]	0.02 [0.58]	0.07 [1.17]	0.07 [1.56]	0.14 [2.51]	0.28 [3.77]
α_{FF5}	-0.17 [-3.29]	-0.03 [-0.81]	0.09 [1.55]	0.01 [0.13]	0.10 [1.89]	0.28 [3.78]
α_{FF6}	-0.15 [-2.85]	-0.02 [-0.42]	0.11 [1.93]	-0.01 [-0.23]	0.10 [1.79]	0.25 [3.39]
Panel B: Fama and French (2018) 6-factor model loadings for LEI-sorted portfolios						
β_{MKT}	1.08 [86.99]	0.98 [98.38]	0.98 [73.95]	0.96 [94.17]	1.03 [80.69]	-0.04 [-2.49]
β_{SMB}	0.11 [5.94]	-0.12 [-8.01]	-0.02 [-1.07]	-0.03 [-1.91]	0.13 [6.92]	0.02 [0.88]
β_{HML}	0.02 [0.92]	0.15 [7.58]	-0.11 [-4.31]	-0.01 [-0.52]	-0.10 [-4.12]	-0.12 [-3.74]
β_{RMW}	0.10 [4.13]	0.10 [5.30]	-0.05 [-2.05]	0.11 [5.69]	0.04 [1.41]	-0.06 [-1.93]
β_{CMA}	-0.12 [-3.33]	0.02 [0.65]	-0.13 [-3.46]	0.21 [7.16]	0.15 [4.01]	0.27 [5.39]
β_{UMD}	-0.04 [-2.94]	-0.03 [-2.65]	-0.04 [-2.68]	0.02 [2.45]	0.01 [0.56]	0.04 [2.54]
Panel C: Average number of firms (n) and market capitalization (me)						
n	674	526	1062	584	648	
me (\$10 ⁶)	1494	3076	2226	3098	1458	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LEI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.73]	0.32 [4.33]	0.32 [4.40]	0.28 [3.77]	0.28 [3.78]	0.25 [3.39]
Quintile	NYSE	EW	0.18 [3.99]	0.21 [4.58]	0.20 [4.29]	0.18 [3.78]	0.19 [3.97]	0.17 [3.63]
Quintile	Name	VW	0.26 [3.71]	0.32 [4.57]	0.33 [4.74]	0.28 [3.97]	0.27 [3.92]	0.24 [3.44]
Quintile	Cap	VW	0.23 [3.48]	0.27 [4.17]	0.28 [4.22]	0.23 [3.43]	0.19 [2.99]	0.16 [2.50]
Decile	NYSE	VW	0.31 [3.38]	0.36 [3.92]	0.35 [3.79]	0.36 [3.81]	0.23 [2.47]	0.25 [2.70]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.22 [2.92]	0.25 [3.42]	0.25 [3.46]	0.23 [3.13]	0.22 [2.97]	0.20 [2.72]
Quintile	NYSE	EW	-0.08 [-1.29]					
Quintile	Name	VW	0.21 [2.86]	0.26 [3.76]	0.27 [3.89]	0.25 [3.50]	0.23 [3.26]	0.20 [2.97]
Quintile	Cap	VW	0.18 [2.69]	0.22 [3.43]	0.23 [3.45]	0.20 [3.04]	0.16 [2.44]	0.14 [2.12]
Decile	NYSE	VW	0.24 [2.57]	0.29 [3.09]	0.28 [2.98]	0.29 [3.02]	0.18 [1.92]	0.18 [1.97]

Table 3: Conditional sort on size and LEI

This table presents results for conditional double sorts on size and LEI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LEI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LEI and short stocks with low LEI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LEI Quintiles					LEI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.67 [2.46]	0.91 [3.30]	0.89 [3.30]	0.88 [3.19]	0.80 [2.87]	0.13 [1.48]	0.15 [1.72]	0.14 [1.60]	0.09 [1.03]	0.11 [1.27]	0.08 [0.91]
	(2)	0.71 [2.66]	0.90 [3.62]	0.84 [3.31]	0.87 [3.53]	0.87 [3.44]	0.16 [2.00]	0.20 [2.47]	0.17 [2.20]	0.17 [2.14]	0.13 [1.70]	0.14 [1.72]
	(3)	0.80 [3.17]	0.79 [3.60]	0.82 [3.42]	0.82 [3.66]	0.88 [3.75]	0.09 [1.09]	0.14 [1.85]	0.13 [1.68]	0.09 [1.09]	0.12 [1.45]	0.08 [1.04]
	(4)	0.72 [3.11]	0.81 [3.92]	0.82 [3.67]	0.73 [3.52]	0.92 [4.24]	0.20 [2.63]	0.24 [3.22]	0.23 [3.07]	0.19 [2.46]	0.21 [2.66]	0.18 [2.23]
	(5)	0.47 [2.38]	0.63 [3.50]	0.59 [3.06]	0.60 [3.31]	0.83 [4.33]	0.36 [4.01]	0.38 [4.24]	0.39 [4.36]	0.33 [3.66]	0.32 [3.52]	0.28 [3.09]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LEI Quintiles					LEI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	392	393	393	393	388	37	34	32	33	35	
	(2)	107	107	107	107	107	61	62	60	61	61	
	(3)	76	77	76	77	76	107	110	105	106	108	
	(4)	64	65	64	65	64	229	241	229	234	231	
(5)	59	59	59	59	59	1452	2217	1819	2183	1506		

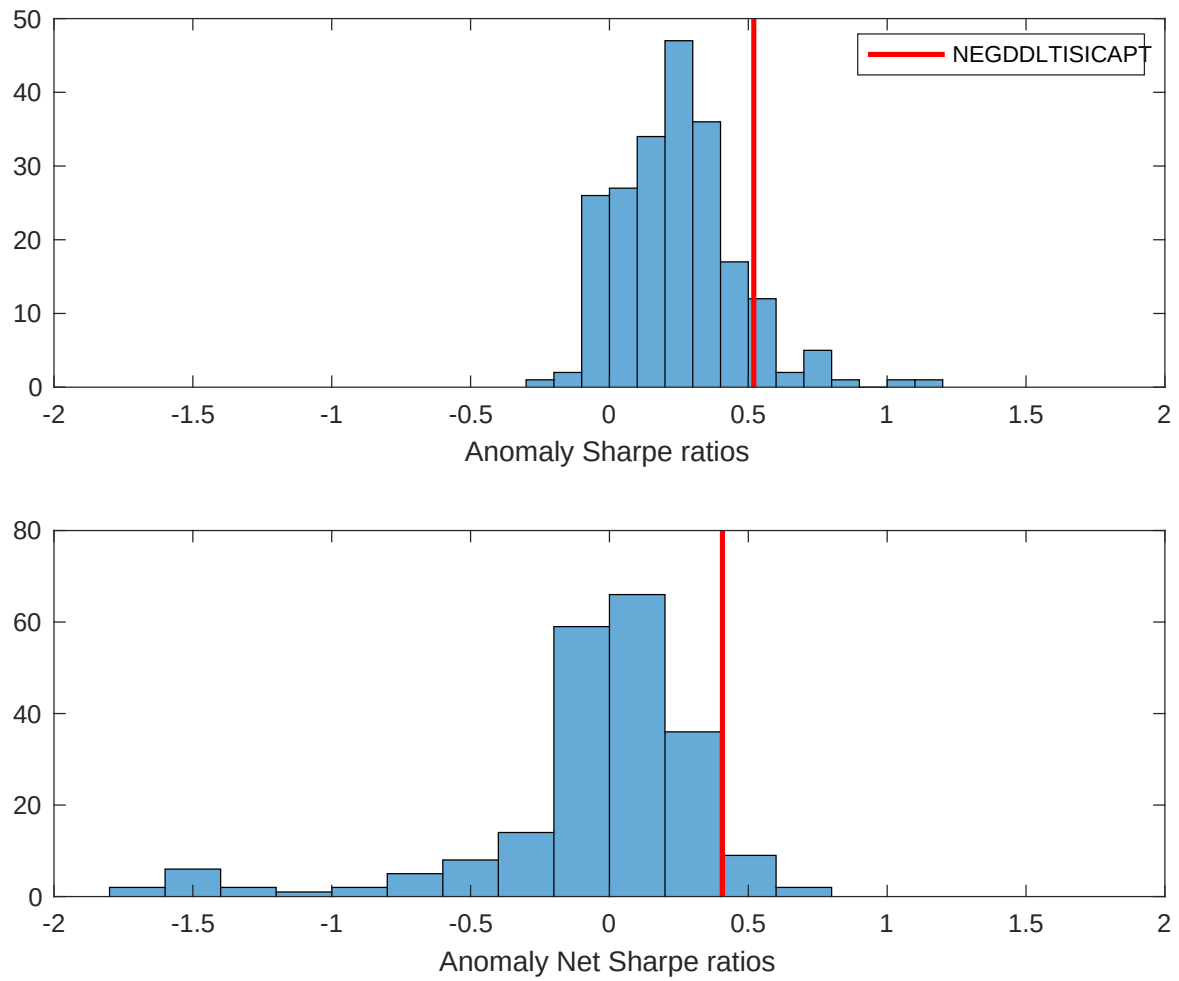


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LEI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

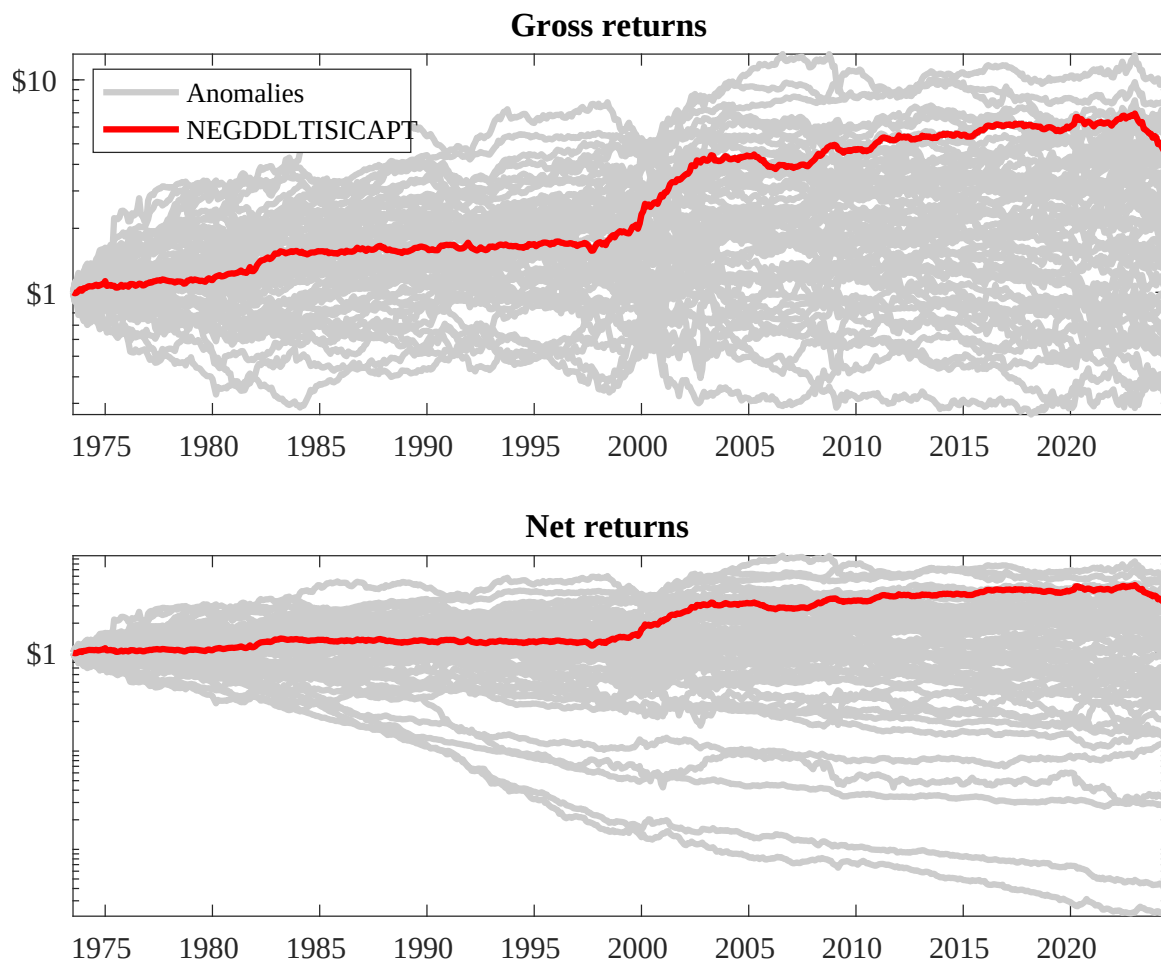
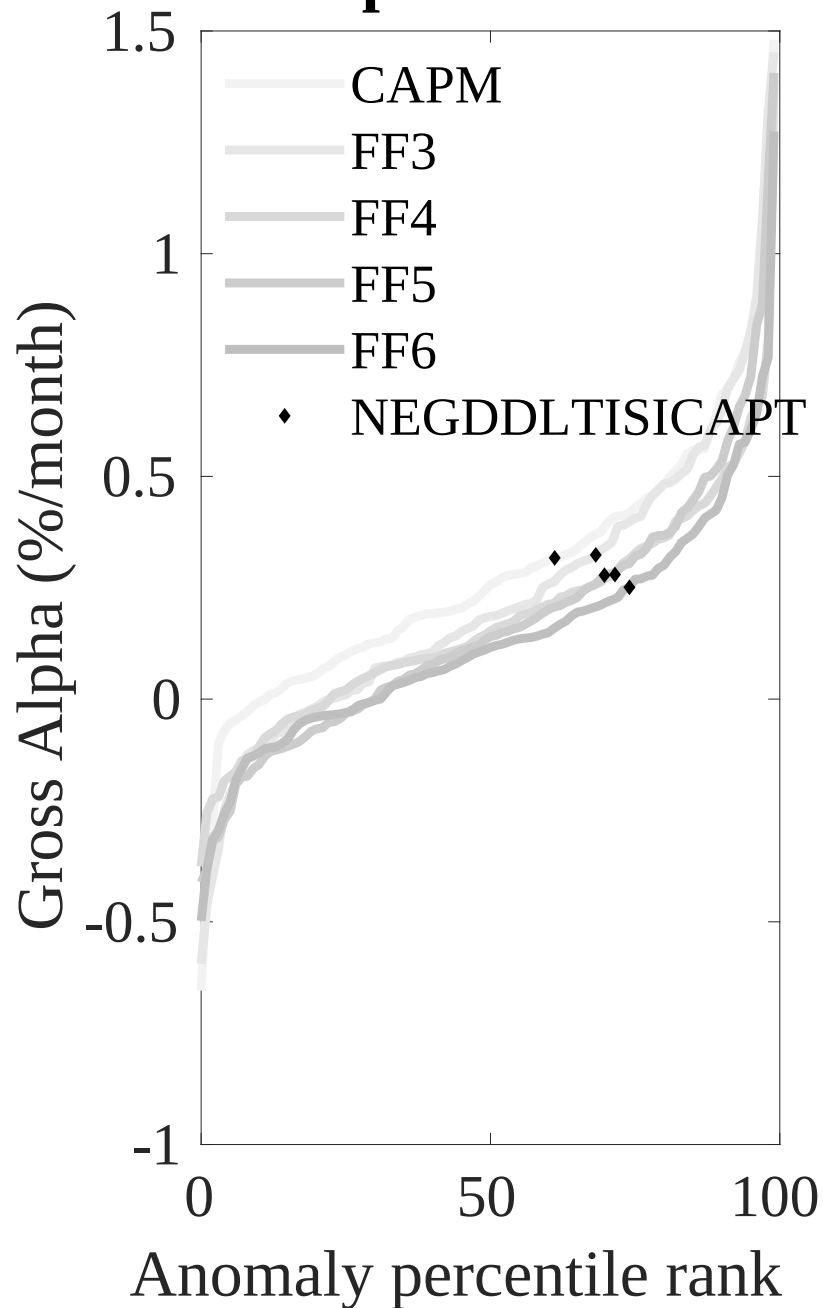


Figure 3: Dollar invested.

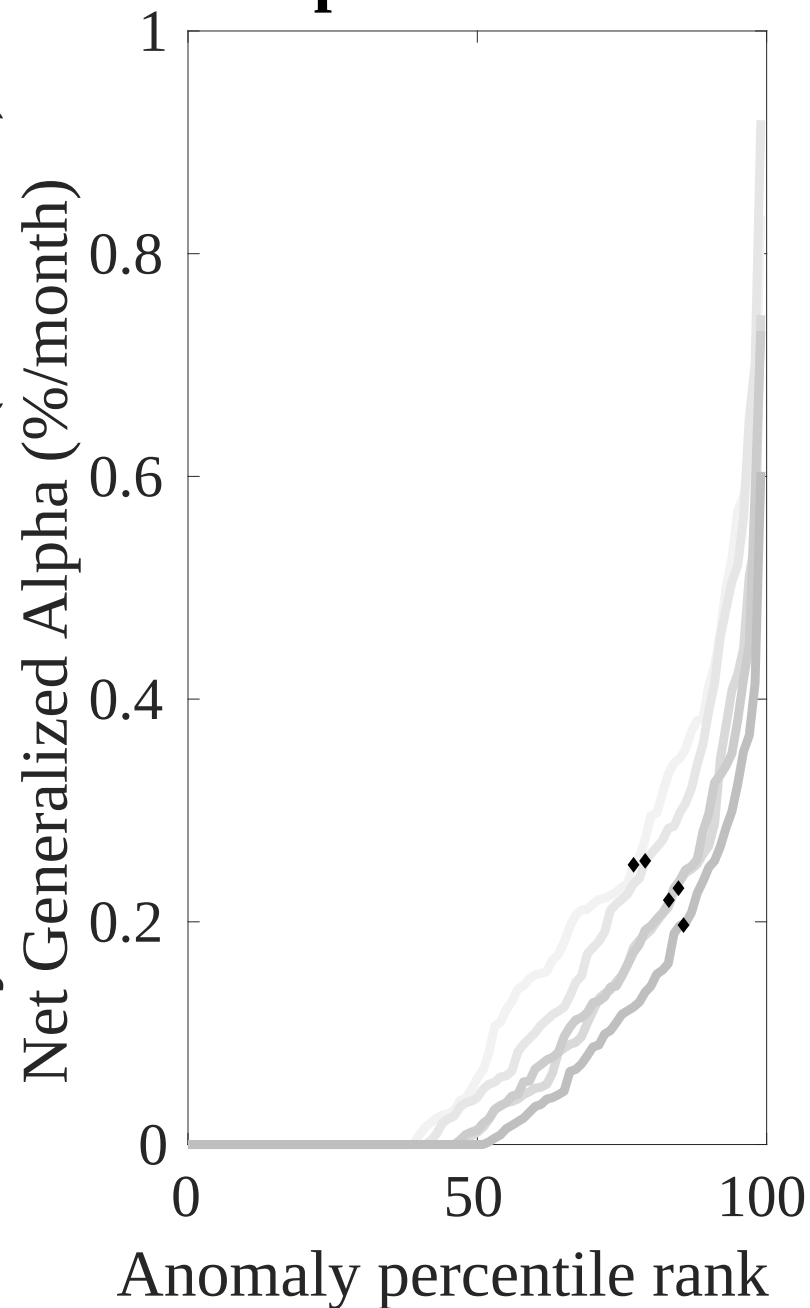
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LEI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



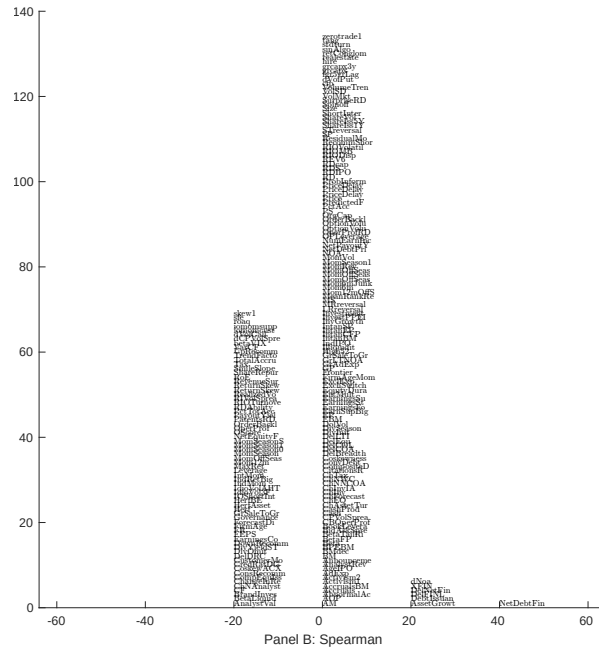
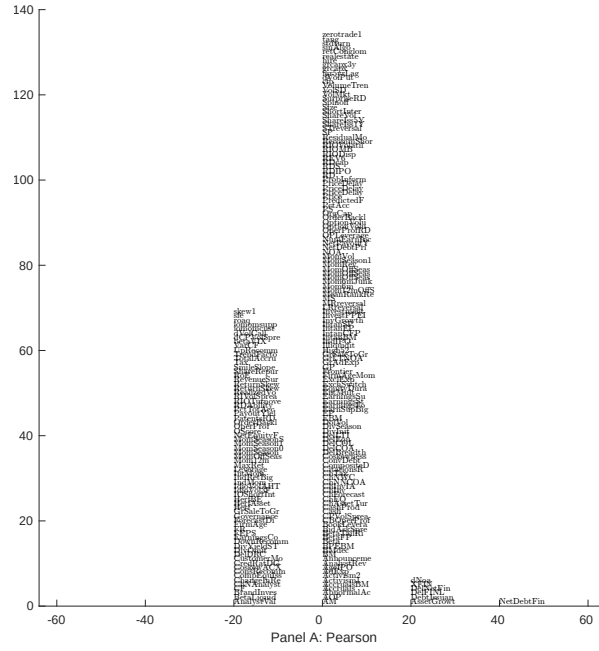


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with LEI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

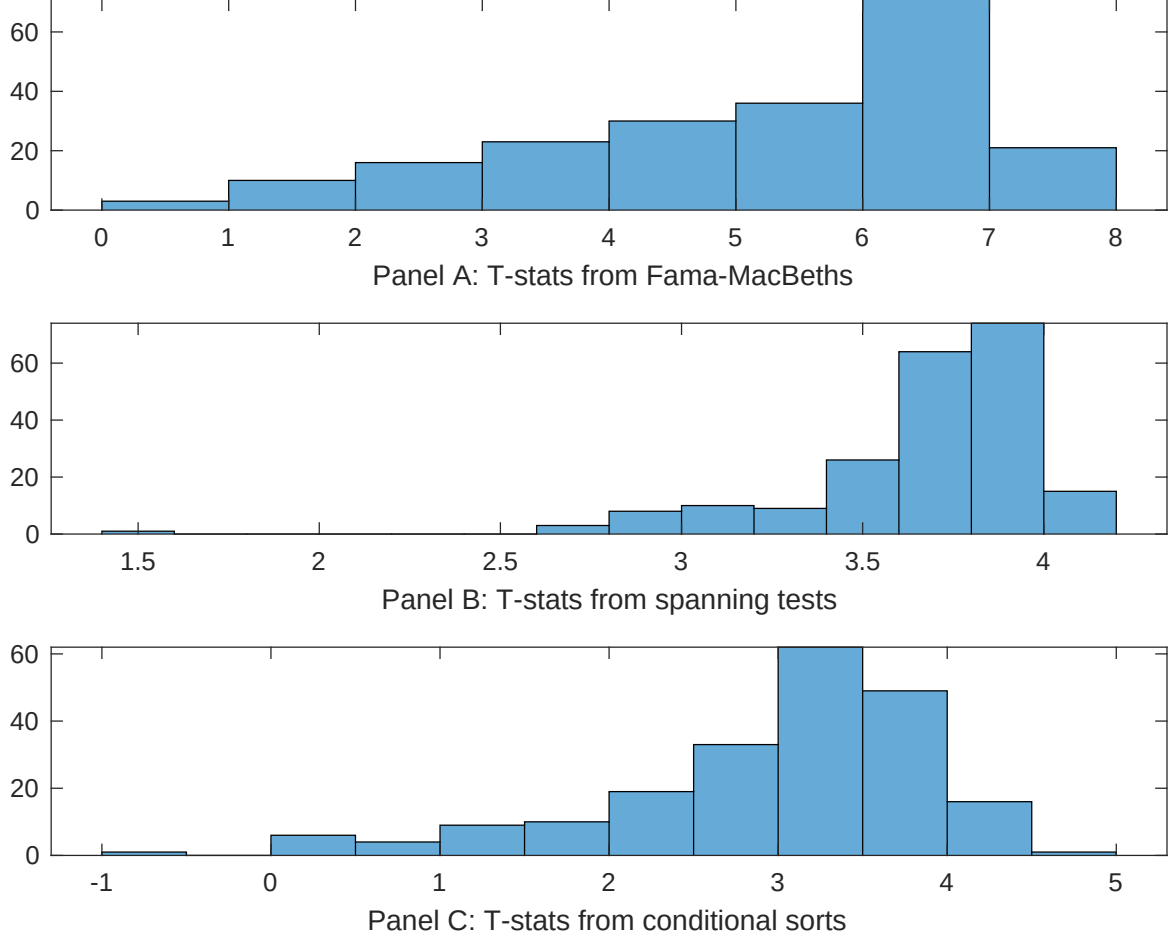


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LEI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LEI}LEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LEI. Stocks are finally grouped into five LEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LEI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LEI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LEI}LEI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Inventory Growth, change in ppe and inv/assets, Net external financing, Change in net financial assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.37]	0.13 [5.33]	0.14 [5.32]	0.14 [5.74]	0.14 [5.69]	0.13 [5.25]	0.15 [5.72]
LEI	0.89 [1.50]	0.10 [1.58]	0.34 [5.10]	0.16 [2.54]	0.16 [2.35]	0.29 [4.66]	0.11 [0.14]
Anomaly 1	0.17 [9.07]						-0.24 [-0.05]
Anomaly 2		0.20 [8.42]					0.86 [1.29]
Anomaly 3			0.40 [6.86]				0.69 [1.19]
Anomaly 4				0.17 [7.62]			0.13 [4.10]
Anomaly 5					0.19 [6.09]		0.13 [2.36]
Anomaly 6						0.61 [4.03]	-0.52 [-1.50]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LEI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LEI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Inventory Growth, change in ppe and inv/assets, Net external financing, Change in net financial assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.22 [3.00]	0.22 [3.07]	0.24 [3.32]	0.24 [3.26]	0.22 [3.02]	0.20 [2.78]	0.19 [2.67]
Anomaly 1	18.41 [4.47]						2.01 [0.35]
Anomaly 2		21.09 [5.34]					10.96 [2.05]
Anomaly 3			8.03 [2.81]				7.25 [2.48]
Anomaly 4				7.03 [2.12]			-1.94 [-0.55]
Anomaly 5					14.86 [4.07]		11.33 [2.98]
Anomaly 6						17.72 [5.17]	13.32 [3.44]
mkt	-3.54 [-2.11]	-3.92 [-2.36]	-4.11 [-2.43]	-4.06 [-2.40]	-1.92 [-1.10]	-3.63 [-2.18]	-2.17 [-1.26]
smb	0.61 [0.24]	0.63 [0.25]	3.38 [1.31]	2.52 [0.99]	7.27 [2.60]	3.53 [1.40]	6.51 [2.26]
hml	-10.91 [-3.37]	-11.28 [-3.52]	-12.39 [-3.80]	-12.88 [-3.92]	-10.48 [-3.22]	-12.45 [-3.88]	-10.61 [-3.30]
rmw	-7.89 [-2.40]	-8.55 [-2.61]	-5.07 [-1.52]	-6.11 [-1.84]	-15.12 [-3.83]	-4.11 [-1.25]	-11.78 [-2.98]
cma	21.24 [4.18]	21.94 [4.41]	20.26 [3.65]	21.83 [3.89]	17.71 [3.25]	31.93 [6.45]	15.06 [2.37]
umd	2.91 [1.71]	3.14 [1.88]	3.68 [2.16]	4.49 [2.67]	4.40 [2.64]	4.20 [2.54]	2.56 [1.52]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	12	14	11	10	12	13	17

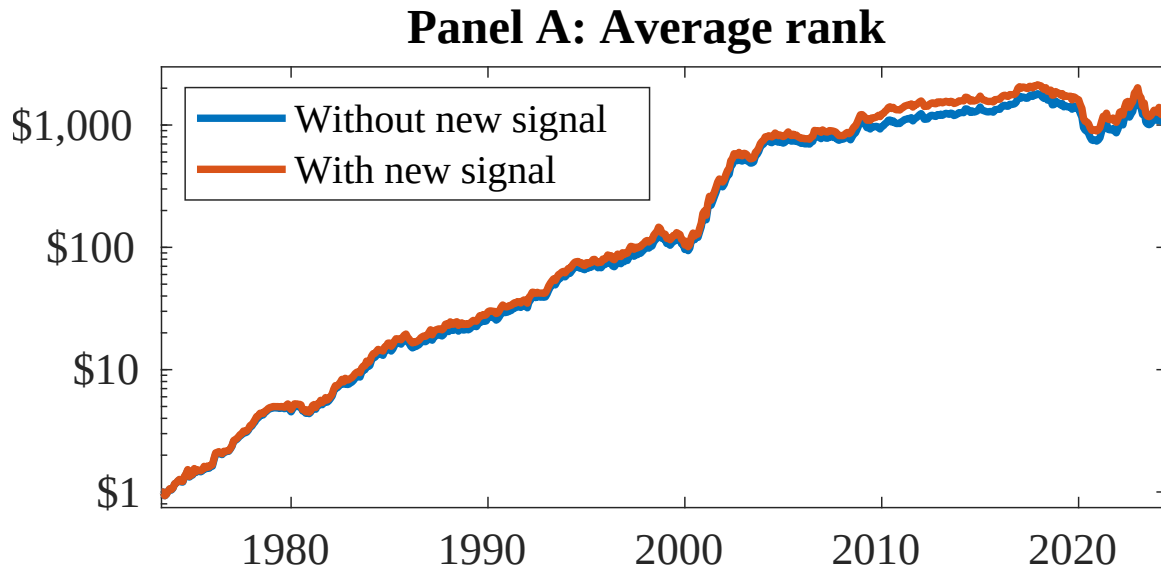


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LEI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Berk, J. B., Green, R. C., and Naik, V. (1999). Optimal investment, growth options, and security returns. *Journal of Finance*, 54(5):1553–1607.
- Breeden, D. T. (1979). An intertemporal asset pricing model with stochastic consumption and investment opportunities. *Journal of Financial Economics*, 7(3):265–296.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H., Piazzesi, M., and Hansen, L. P. (2008). Asset pricing explorations for macroeconomics. *NBER Macroeconomics Annual*, 23:115–165.
- Daniel, K. and Titman, S. (2012). Testing factor-model explanations of market anomalies. *Critical Finance Review*, 1(1):103–139.

- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, *Forthcoming*.
- Fama, E. F. and French, K. R. (1992). The cross-section of expected stock returns. *Journal of Finance*, 47(2):427–465.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.
- George, T. J. and Hwang, C.-Y. (2010). A resolution of the distress risk and leverage puzzles in the cross section of stock returns. *Journal of Financial Economics*, 96(1):56–79.
- Gomes, J., Kogan, L., and Zhang, L. (2003). Equilibrium cross section of returns. *Journal of Political Economy*, 111(4):693–732.
- Hou, K., Xue, C., and Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5):2019–2133.
- Kozak, S., Nagel, S., and Santosh, S. (2018). Interpreting factor models. *Journal of Finance*, 73(3):1183–1223.
- McLean, R. D. and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.